

SOMA 1

A National Digital Curriculum Platform for Rwanda

Every primary learner, every lesson, every day.

Designed for Rwanda's **3,831** primary schools

\$1.80 per student per year

Sole author: concept note, system architecture, cost model

Submitted for ministerial review — not yet adopted



Context and the *aha*

Rwanda: **92.8%** net primary attendance,
99.8% of teachers formally qualified

Primary 3 English literacy ranges from **32%**
in Kigali to **4%** in Eastern Province — an **8x**
gap on the same national curriculum and
timetable

Kenya's FLANA 2023 data shows the same
pattern elsewhere: a full curriculum redesign
(CBC) lifted numeracy while reading
still declined

The gap tracks **geography**, not ability,
or curriculum design



Hypothesis and validation approach

HYPOTHESIS

Student performance is proportional to the quality of teaching a student is exposed to

63,046

TEACHERS

across 3,831 schools

Not achievable within one strategic-plan cycle through training alone



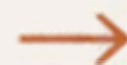
DESIGNED WITH

The Ministry of Education and NESA



PILOT SCOPE

3 recently underperforming schools



EVALUATION DESIGN

Matched treatment and control groups

300+ pupils



DELIVERY MODEL

Solar-powered delivery

20-30 curated videos on the topics students most consistently fail

Evidence

STRONG DIAGNOSTIC EVIDENCE

8X

literacy gap
between Kigali
and Eastern Province

23:1

pupil-teacher ratio
in Kigali

~70:1

pupil-teacher ratio in Kirche

PILOT STATUS

Feasibility pilot
is designed,
not yet run.

Validation is upcoming,
pending Ministry sign-off.



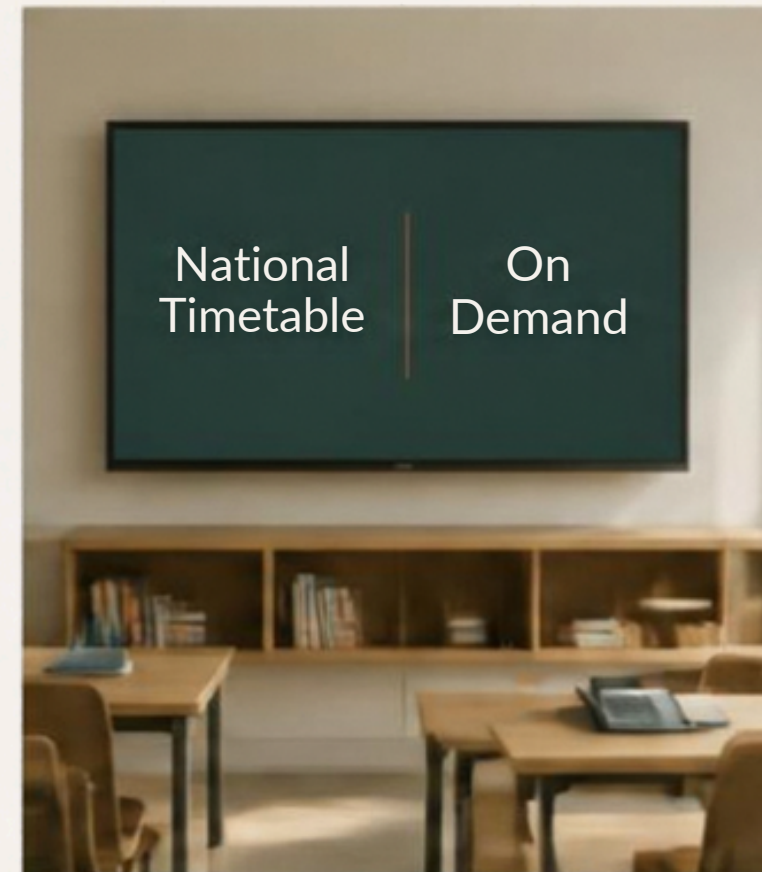
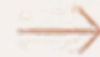
The concept

40-60

master teachers
boot camp

800-1,000

lessons library
grades 1-6



\$3.77M

\$1.80

per student
per year

Master Teacher Boot Camp builds an 800-1,000 lesson library across Grades 1-6. Content is validated by a joint REB/NESA/University of Rwanda board.

Lessons are delivered live on the national timetable and on demand. Classroom teachers repositioned as facilitators.

Solar-powered screens ensure access in the roughly 20% of schools still without electricity.

Full national deployment costs at approximately \$3.77M in Year 1 — about \$1.80 per student per year.

Outcome and role

Sole author of the concept note, system architecture, cost model and ministerial submission

Solar-infrastructure partnership secured with ENGIE Energy Rwanda; scoping visits completed in Kenya and Uganda

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Lessons

THE RISK

hardware without sustainability

Kenya's Digital Literacy Programme shows the real risk in hardware-led EdTech is unfunded maintenance, not device procurement

85%+ urban device usage versus **23%** rural, nearly a decade on

SOMA 1 treats maintenance and climate fit as **core design**, not an afterthought.

THE DESIGN CHOICE

sustainability by design

SOMA 1 budgets maintenance and climate-specific hardware from day one

planned for durability, built for impact.

